

Chronic Stable Angina: Anti-Anginal Drug Pharmacology



1. Nitrate-free interval

WHAT YOU SEE

Clock '10–12 hour daytime-on / patch off at night'

WHAT IT ENCODES

Nitrate tolerance develops with continuous exposure, so a daily nitrate-free interval of 10–12 hours (e.g., remove transdermal NTG patch at night) is mandatory to restore efficacy. Long-acting nitrates reduce preload via venodilation.

BOARD TRAP

24-hour continuous nitrate dosing — causes tachyphylaxis; the nitrate-free interval is the discriminator that preserves efficacy.

2. Transdermal NTG patch / tolerance knight

WHAT YOU SEE

Nitrate Knight, transdermal patch, 'tolerance'

WHAT IT ENCODES

Long-acting nitrates (transdermal NTG, isosorbide mononitrate/dinitrate) prevent angina but cause tolerance without a drug-free interval. They work by venodilation lowering LV preload and myocardial O₂ demand.

BOARD TRAP

Using long-acting nitrate for acute attacks — they are prophylactic; sublingual NTG is for acute symptoms.

3. Sublingual NTG for acute attack

WHAT YOU SEE

Sublingual tablet knight, 'taken at onset'

WHAT IT ENCODES

Sublingual nitroglycerin is the rescue agent for acute angina — taken at symptom onset, may repeat every 5 minutes up to 3 doses. If pain persists after 3 doses / 15 minutes, seek emergency care (possible ACS).

BOARD TRAP

Relief with SL NTG 'proves' cardiac angina — esophageal spasm also responds, so response is not diagnostic of CAD.

4. Avoid short-acting nifedipine

WHAT YOU SEE

Red 'IR NIFEDIPINE — precipitates MI'

WHAT IT ENCODES

Immediate-release (short-acting) dihydropyridine nifedipine causes reflex tachycardia and sympathetic activation that can precipitate MI — avoid in angina/ACS. Use long-acting DHPs (e.g., amlodipine) instead.

BOARD TRAP

Short-acting nifedipine for angina/HTN urgency — reflex tachycardia worsens ischemia; long-acting DHP is safe.

5. Amlodipine — long-acting DHP, safe with BB

WHAT YOU SEE

Amlodipine 'DHP beauty, safe with BB'

WHAT IT ENCODES

Amlodipine is a long-acting dihydropyridine CCB — peripheral vasodilator, safe to combine with a beta-blocker, and does not depress AV node. Good add-on when beta-blocker alone fails to control angina.

BOARD TRAP

Worrying about bradycardia from amlodipine + BB — DHPs don't block AV node; that danger belongs to verapamil/diltiazem.

6. Verapamil/diltiazem — AV block, avoid with BB

WHAT YOU SEE

Verapamil Knight, X over Beta-Blocker soldier, 'AV block/bradycardia/HF'

WHAT IT ENCODES

Non-dihydropyridine CCBs (verapamil, diltiazem) slow the AV node and depress contractility. Combining them with a beta-blocker risks severe bradycardia, heart block, and decompensated HF. Avoid in reduced EF.

BOARD TRAP

Adding verapamil to a beta-blocker — additive AV nodal blockade; the safe CCB partner for a BB is a DHP like amlodipine.

7. Beta-blocker — never stop abruptly

WHAT YOU SEE

Beta-Blocker soldier, 'abrupt stop → rebound, 2-week taper'

WHAT IT ENCODES

Beta-blockers are first-line anti-anginals (reduce HR, contractility, O₂ demand). Abrupt withdrawal causes rebound tachycardia and increased myocardial O₂ demand — risk of ischemia/MI. Taper over ~2 weeks.

BOARD TRAP

Stopping a beta-blocker abruptly in CAD — rebound adrenergic surge can trigger MI; always taper.

8. Ranolazine — late Na current, QT

WHAT YOU SEE

Ranolazine flask & pharmacist, 'QT prolongation, no HR/BP effect'

WHAT IT ENCODES

Ranolazine inhibits the late inward sodium current, reducing calcium overload and wall tension without changing HR or BP. Add-on for refractory angina. It prolongs the QT interval and levels rise with CYP3A inhibitors.

BOARD TRAP

Expecting ranolazine to lower HR or BP — it does neither; its hemodynamic-neutral profile and QT prolongation are the discriminators.

9. Ivabradine — funny current, needs sinus rhythm

WHAT YOU SEE

Ivabradine, 'sinus node (If), only if EF reduced & HR $\geq$ 70 in SR'

WHAT IT ENCODES

Ivabradine selectively inhibits the If 'funny' current in the SA node, lowering heart rate without affecting BP or contractility. Used in stable angina/HFrEF with sinus rhythm and resting HR $\geq$ 70 despite max beta-blocker.

BOARD TRAP

Giving ivabradine in atrial fibrillation — it only works in sinus rhythm via the SA node and is ineffective in AF.

10. Avoid NSAIDs

WHAT YOU SEE

Red X over NSAIDs

WHAT IT ENCODES

NSAIDs cause sodium/fluid retention, raise BP, blunt antihypertensives, and increase cardiovascular/thrombotic risk — avoid in CAD/HF. They also undermine the cardioprotective antiplatelet effect of aspirin if co-administered.

BOARD TRAP

Ibuprofen with low-dose aspirin in CAD — ibuprofen competitively blocks aspirin's irreversible COX-1 platelet inhibition.

11. Drug interaction danger zone — CYP3A

WHAT YOU SEE

Interaction Danger Zone bench, CYP3A bumper

WHAT IT ENCODES

Ranolazine and several CCBs are CYP3A4 substrates; strong CYP3A inhibitors (azoles, macrolides, grapefruit) raise levels and toxicity (e.g., ranolazine-induced QT prolongation). Check interactions before adding anti-anginals.

BOARD TRAP

Co-prescribing a strong CYP3A inhibitor with ranolazine — markedly raises ranolazine and QT-prolongation risk.
